

# Automated Trading Systems for Futures Prop-Firm Evaluations:

## Design Constraints, Risk Architecture, and Detection Countermeasures

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### Abstract

This paper examines the technical and structural feasibility of using algorithmic trading systems ("bots" or Expert Advisors) to complete futures proprietary-trading firm evaluations (e.g., Apex Trader Funding, TopStep, Tradeify, Bulenox). It reviews evaluation rule structures common across the industry, the risk-management architecture a bot must implement to survive them, the behavioral signatures prop firms use to detect rule-gaming or exploit-based bots, and the legal/contractual standing of automation under current firm policies. A case framework is proposed using an ICT/SMC (Inner Circle Trader / Smart Money Concepts) bias-detection model applied to NAS100, consistent with a discretionary-to-systematic conversion approach. The central finding is that automation can reliably pass an evaluation only when it encodes genuine statistical edge and firm-compliant risk controls — approaches built purely to exploit evaluation mechanics face rising detection risk and rule changes.

### 1. Introduction

Futures prop-firm evaluations (also called "combines" or "challenges") allow traders to demonstrate risk-managed profitability on a simulated or firm-owned account before receiving a funded, profit-split account. Because evaluations are rule-bound and repeatable, they are a natural target for automation: a well-specified bot can execute a strategy with zero emotional deviation, consistent position sizing, and exact rule adherence — traits that are difficult for discretionary traders to maintain.

The premise of "a bot to pass the evaluation" is therefore not inherently illegitimate. Firms generally permit algorithmic execution; what they restrict is exploit-based automation that games the evaluation's statistical structure rather than trading a real edge. This paper separates the two and details what a compliant, evaluation-ready bot requires.

### 2. Evaluation Rule Structures

Across major futures prop firms, evaluation rules cluster around six constraint types. Table 1 summarizes these and the corresponding requirement they place on bot architecture.

Table 1. Common Evaluation Constraints and Bot Design Implications

| Constraint            | Typical Rule                                                          | Bot Design Implication                                                                                        |
|-----------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Max Daily Loss        | 3–5% of starting/EOD balance                                          | Bot must halt new entries once cumulative daily P&L breaches threshold; kill-switch, not just position sizing |
| Max Overall Drawdown  | 6–10%, trailing or static                                             | Equity curve monitored intra-trade; trailing DD needs high-water-mark tracking, not just balance-based DD     |
| Profit Target         | 8–10% (Phase 1), 5% (Phase 2)                                         | Bot needs a stopping condition — most eval failures are from bots that keep trading after target is hit       |
| Minimum Trading Days  | 5–10 days                                                             | Bot must spread activity across sessions, not front-load risk into one day                                    |
| Consistency Rule      | No single day > 20–40% of total profit                                | Position sizing must be capped per day even if a high-probability setup appears                               |
| Prohibited Strategies | HFT, latency arbitrage, copy trading, martingale/grid (firm-specific) | Strategy logic must be auditable and explainable — avoid black-box martingale scaling                         |

### 3. Why Naive Bots Fail Evaluations

Three failure modes account for most automated evaluation failures observed in retail trading communities and firm post-mortems:

1. Over-optimization to historical data (curve-fitting): a bot backtested to a specific NAS100 or ES session window performs well historically but has no genuine edge, producing a random-walk outcome live.
2. No daily-loss kill switch: the bot's position-sizing logic caps risk per trade but not cumulative daily risk, so a losing streak within one session breaches the daily-loss rule even though each individual trade was "small."

3. Ignoring the consistency rule: a bot that has one outsized winning day (common with breakout or news-driven strategies) can technically pass the profit target while violating the requirement that no single day account for a large share of total profit — this is the single most common reason automated passes are later rejected at the funded stage.

## 4. Required Architecture for a Compliant Evaluation Bot

### 4.1 Risk Layer (must sit above strategy logic)

- Daily loss governor: tracks running P&L against starting-of-day balance; disables new order submission once a configurable buffer below the firm's daily-loss limit is reached (e.g., stop at 80% of the allowed daily loss, not 100%).
- Trailing/static drawdown governor: separately tracks high-water-mark equity for firms using trailing drawdown; this is a different calculation from daily loss and is the more common source of unexpected breaches.
- Profit-target lock: once the evaluation profit target is reached, the bot should reduce size or halt rather than continue compounding risk — most rule sets do not require stopping, but continuing raises drawdown-breach risk for no additional benefit.
- Per-trade and per-day risk caps sized so that no single day can mathematically exceed the consistency-rule ceiling, even in a best-case outcome.

### 4.2 Strategy Layer

The strategy layer is what separates a legitimate systematic trader from a rule-exploiter. A defensible strategy layer should be:

- Statistically testable out-of-sample (walk-forward, not just in-sample backtest) — consistent with the Java-based EUR/USD walk-forward/Monte Carlo framework referenced in your prior EA work.
- Session- and structure-aware rather than purely indicator-driven — e.g., an ICT/SMC bias model using CHoCH/BOS confirmation, FVG/OB entries, and liquidity-sweep (SSL/BSL) filters gives each trade an explainable structural reason, which matters both for firm audits and for genuine edge.
- Regime-aware: many evaluation failures come from a single strategy applied across incompatible regimes (trending vs. range). A bias-detection layer that classifies the session before allowing entries (as in a daily NAS100 bias framework) reduces this risk.

### 4.3 Execution & Compliance Layer

- Explicit trading-day counter to satisfy minimum-trading-day requirements without forcing trades on low-quality days.
- Logging of every decision variable (structure state, session, risk used) — firms increasingly request logs or MT5/NinjaTrader journal exports during funded-stage review.
- No latency-arbitrage, no tick-scalping below the firm's stated minimum hold time, no martingale/grid position scaling — these are explicitly banned by most firms (Apex, TopStep, and others publish this in their trading rules) and are the primary basis for evaluation disqualification even after a nominal "pass."

## 5. Detection Methods Used by Prop Firms

Firms increasingly run automated statistical review on evaluation accounts before funding a trader. Reported detection signals include: abnormal win rate with no drawdown variance (suggesting server-side exploits or copy-trading), trade clustering at identical millisecond intervals (bot signature without risk variance), single-day profit concentration inconsistent with the consistency rule, and hold times below stated minimums. A bot designed around genuine strategy logic with human-plausible variance in entry timing and position sizing is far less likely to trigger these reviews than a bot optimized purely to hit the numeric pass threshold fastest.

## 6. Case Framework: NAS100 ICT/SMC Bias Bot

Given an existing daily NAS100 intraday bias workflow using ICT/SMC methodology (CHoCH, BOS, FVG, OB, SSL/BSL, PDRA, OTE, Judas swing) and a 10-step manual checklist, the natural conversion path to an evaluation-ready bot is:

1. Encode the 10-step checklist as discrete boolean/scored conditions (structure shift confirmed, liquidity swept, FVG/OB present, session filter, macro-news filter).
2. Backtest and walk-forward validate on NAS100 futures (MNQ) tick or 1-minute data across multiple regimes, not a single trending period.
3. Layer the risk governor (Section 4.1) on top — this layer is firm-specific and should be reconfigurable per prop firm's exact rule set (Apex vs. TopStep vs. Tradeify differ in trailing-DD calculation).
4. Paper-trade or sim-forward test for at least one full evaluation cycle before live evaluation deployment, tracking whether the consistency rule would have been breached on any single day.

This mirrors the structure of the MQL5 EAs already built for XAUUSD/EURUSD (NY-session breakout, EMA crossover, RSI filter, ATR-based sizing, 1%-daily-loss snapshot) — the same risk-management pattern applies directly to a futures/NAS100 evaluation bot, with the addition of a trailing-drawdown governor specific to futures prop-firm rules.

## 7. Legal, Contractual, and Ethical Considerations

- Most firms explicitly permit EAs/algorithmic execution in their terms of service — verify per firm, as policies change and some restrict specific server-side automation or require disclosure.
- Exploiting mechanical loopholes (e.g., hedging across multiple evaluation accounts to guarantee a pass, or using tick-scalping strategies below stated minimum hold times) typically violates terms of service even if not technically "illegal," and firms have broadly discretionary rights to revoke funded accounts for this.
- A bot trading a genuine, validated edge with firm-compliant risk controls is functionally no different from firm-permitted discretionary automation — the ethical and contractual risk is concentrated in exploit-oriented designs, not in automation itself.

## 8. Conclusion

A bot can pass a futures prop-firm evaluation, and automation is generally permitted by the firms themselves. The determining factor for durable success — passing evaluation and retaining a funded account — is whether the bot encodes a validated strategy with firm-compliant risk governance (daily-loss kill switch, trailing-drawdown tracking, consistency-rule-aware position sizing) rather than a purely rule-exploiting design. The existing ICT/SMC NAS100 bias framework and prior MQL5/Java EA risk architecture provide a direct, reusable foundation for building such a system.

## References

*[1] Prop firm rule documentation should be sourced directly from each firm's current published trading rules (Apex Trader Funding, TopStep, Tradeify, Bulenox), as thresholds and permitted-strategy lists change frequently and vary by firm.*

*[2] ICT/SMC methodology concepts (CHoCH, BOS, FVG, OB, liquidity sweeps) as applied in the author's existing daily NAS100 bias workflow.*

*[3] Prior Expert Advisor risk-management patterns (NY-session breakout, ATR sizing, daily-loss snapshot) developed for XAUUSD/EURUSD in MQL5.*